

ReMynd Assessment Operating System (RAOS) – Standalone Platform

OBJECTIVE

Design and build a **secure, scalable, standalone web-based Assessment Operating System** for psychoeducational services.

The system must manage the **full lifecycle of a student case**:

- Parent onboarding (consent + intake)
- AI-assisted assessment selection
- Multi-respondent assessment deployment (QR + link)
- Guided self-report administration
- Automated scoring and aggregation
- AI-generated professional reports
- Case progress monitoring (21–30 day workflow)
- Optional export to external intervention systems

This system must be:

- ✓ Role-based
- ✓ Highly visual (dashboards)
- ✓ Low-friction for parents and teachers
- ✓ Designed for international school environments
- ✓ Non-diagnostic (screening + decision-support)

CORE SYSTEM ARCHITECTURE

1. AUTHENTICATION & ROLES

Roles:

- **Admin (System Owner / Noel)**
 - Full access
 - Case creation, tool selection, report approval

- **Assessment Lead (Hayley)**
 - Parent communication
 - Self-report administration (live sessions)
 - Observation input
 - Debrief coordination
- **Psychometrician (Abegail)**
 - Scoring
 - Data validation
- **External Respondents (No login required)**
 - Parent
 - Teacher 1
 - Teacher 2
 - Student

Access via secure tokenized links (QR + URL)

DATABASE SCHEMA (RELATIONAL)

Case Table

```
Case {
  id,
  student_name,
  dob,
  school,
  language_preference,
  case_status,
  current_phase,
  progress_percentage,
  created_at,
  updated_at
}
```

User Table

```
User {
  id,
  name,
  role,
  email,
  assigned_cases[]
}
```

Assessment Tool Table

```
AssessmentTool {  
  id,  
  name,  
  category,  
  respondent_types,  
  is_remynd_owned,  
  scoring_type (auto/manual),  
  domains[]  
}
```

Assignment Table

```
Assignment {  
  id,  
  case_id,  
  tool_id,  
  respondent_type,  
  respondent_label (Parent / Teacher1 / Teacher2 / Student),  
  assigned_to_name,  
  assigned_to_email,  
  unique_link,  
  qr_code,  
  status,  
  due_date,  
  submitted_at  
}
```

Response Table

```
Response {  
  id,  
  assignment_id,  
  answers[],  
  language,  
  submitted_at  
}
```

Score Table

```
Score {  
  id,  
  case_id,  
  tool_id,  
  respondent_type,
```

```
raw_score,  
domain_scores,  
normalized_scores,  
agreement_index,  
generated_at  
}
```

SYSTEM MODULES

MODULE 1: CASE CREATION

Admin creates case:

- Student details
- Assign internal team
- System generates:
 - Case ID
 - Secure access tokens

MODULE 2: CONSENT + PARENT INTAKE

- Digital forms (mobile-friendly)
- Delivered via:
 - QR code
 - Secure link

MODULE 3: AI INTAKE ANALYSIS

Use AI to:

- Analyze intake responses
- Detect key domains:
 - Attention
 - Emotional regulation
 - Executive function
 - Social communication
 - Academic persistence

Output:

```
{  
  "recommended_domains": ["attention", "executive_function"],  
  "risk_level": "moderate"  
}
```

MODULE 4: ASSESSMENT BUILDER

Admin selects tools from:

ReMynd-Owned Tools:

- RCS-80 (Core Screener)
- RASR
- RARI
- REFI
- RERMS
- RSCP
- RARPS
- RFII

External / Standardized Tools:

- Conners
- BRIEF
- BASC

Smart Assist Option:

System suggests tools based on intake analysis.

MODULE 5: MULTI-RESPONDENT DEPLOYMENT

RULE:

Teacher forms require:

- ✓ Teacher 1
- ✓ Teacher 2

SYSTEM ACTION:

For each assignment:

- Generate unique:
 - Secure URL
 - QR code

Respondent Access:

- No login required
- Role-specific forms only
- Mobile-first design



MODULE 6: FORM COMPLETION MONITOR

Dashboard View:

| Tool | Parent | T1 | T2 | Student | Status |

Status Types:

- Not Started

- In Progress
- Completed
- Overdue

Automation:

- 48-hour reminder
- 5-day escalation
- Admin alerts



MODULE 7: SELF-REPORT ADMIN INTERFACE

Features:

- “Guided Administration Mode”
- Live data entry
- Screen-share optimized
- Language toggle:
 - English
 - Mandarin
 - Cantonese

Use Cases:

- Face-to-face assessment
- Zoom-based administration



MODULE 8: DATA ROUTING

All responses:

→ Automatically stored in case-specific data structure



MODULE 9: SCORING ENGINE

ReMynd Tools:

- Auto-scored
- Domain-level outputs
- Visualization:
 - Radar charts
 - Bar graphs

Teacher Aggregation:

```
teacher_avg = (T1 + T2) / 2  
agreement_index = Math.abs(T1 - T2)
```

Flags:

- High discrepancy
- Cross-informant mismatch

External Tools:

- Manual score entry
- File upload support



MODULE 10: AI REPORT GENERATOR

Input:

- Intake data
- Scores
- Observations

Output:

Comprehensive Educational Profile & Support Plan

Sections:

- Background summary
- Domain analysis
- Strengths
- Areas of concern
- Cross-setting comparison
- Recommendations



MODULE 11: ADMIN REVIEW PANEL

Admin can:

- Edit report text
- Add notes
- Remove sections



MODULE 12: EXPORT SYSTEM

Formats:

- PDF
- DOCX

MODULE 13: FULL CASE PROGRESS TRACKER

Phases:

1. Pre-Commitment
2. Intake
3. Setup
4. Forms
5. Assessment
6. Scoring
7. Report
8. Debrief
9. Complete

Visual Elements:

- Progress bar
- Phase tracker
- Risk indicators

UI / UX REQUIREMENTS

Admin Dashboard:

- Case overview

- Completion tracking
- Alerts

Role-Based Dashboards:

- Hayley → communication + scheduling
- Abigail → scoring queue

Parent / Teacher Experience:

- Simple
- Mobile-first
- One-click access



SECURITY

- Tokenized access links
- Role-based access control
- Encrypted data storage
- Case-level data isolation



TECH STACK (RECOMMENDED)

- Frontend: React / Next.js
- Backend: Node.js (Express or NestJS)
- Database: PostgreSQL
- Authentication: JWT / Magic Links
- QR Code: `qrcode` library
- Charts: Recharts or Chart.js
- AI Integration: Google AI Studio APIs (Gemini)



AI INTEGRATION REQUIREMENTS

Use Google AI Studio (Gemini) for:

- 1. Intake Analysis**
 - 2. Tool Recommendation**
 - 3. Report Generation**
-

Prompt Structure:

- Structured input (JSON)
 - Output formatted in sections
 - Editable by admin
-



FINAL BUILD INSTRUCTIONS

Build the system as:

- ✓ Modular
- ✓ API-driven
- ✓ Scalable
- ✓ Clean UI
- ✓ Minimal friction for external users

PRIORITIES:

1. Case workflow tracking
2. Multi-respondent form system

3. Scoring + aggregation
4. AI report generation



END PROMPT